The teams have to be formed in group of 2-3 members.

1st week	Team formation n topic finalization
2nd week	Understanding concept and presentation on proposed work
3rd week	Development
4th week	Testing, Evaluation & feedback
5th Week	Jury assessment

MARKING SCHEME FOR JURY ASSESSMENT DURING TA -1 & TA-2

Component	**Marks**
Understanding & Objectives -WRITTEN	2
Design & Architecture – WRITTEN	5
Implementation	5
Performance Analysis	3
Demo & Viva	5
Total	**20**

Evaluation Matrix (40 marks will be mapped to 20 marks as given above)

Criteria	Excellent (4)	Good (3)	Satisfactory (2)	Needs Improvement (1)	Marks
Problem Understanding	Clear understanding of healthcare problem and ML objective	Good understanding with minor gaps	Partial understanding	Problem not clearly defined	5
Dataset Handling & Preprocessing	Proper cleaning, augmentation, feature engineering	Adequate preprocessing	Basic preprocessing	Dataset poorly handled	7
Model Implementation	Correct and efficient implementation of ML/DL model	Model implemented with minor issues	Basic model implementation	Incorrect or incomplete implementation	10
Experimentation & Analysis	Multiple experiments with proper evaluation metrics	Limited experimentation	Minimal analysis	No proper experiments	8

Results & Interpretation	Clear interpretation of results with insights	Results explained moderately	Limited explanation	Results not interpreted	5
Presentation & Demonstration	Clear explanation and working demo	Good explanation with minor issues	Partial clarity	Poor explanation/demo	5

***Extra marks if paper is published (standard journal/conference) based on your mini projects implemented in ta1 & ta2.**

Major Project 2

Federated Learning for Healthcare Time-Series Data (EEG/ECG / Wearable Sensors)

Major Project Problem Statement
Design and implement **a federated learning framework for healthcare time-series data** such as EEG /ECG or wearable sensor signals, addressing challenges of heterogeneous devices, communication efficiency, and distributed model training.
Application Example Datasets
- MIT-BIH Arrhythmia ECG dataset
- CHBMIT EEG dataset
- PhysioNet wearable sensor dataset
- Human Activity Recognition dataset

Mini Project 1
Deep Learning Model for ECG/EEG Signal Classification
Problem Statement
Develop a deep learning model for ECG/EEG signal classification to detect cardiac/cognitive abnormalities using a centralized dataset.
Scope (1 Month)
- ECG/EEG signal preprocessing
- Feature extraction or deep learning model (CNN/LSTM)
- Train and evaluate classification model

Evaluation Metrics
- Accuracy
- F1 Score
- Training time

Expected Outcomes
- Baseline ECG/EEG classification model

Output
✓ Trained ECG/EEG classification model
✓ Signal preprocessing pipeline

Mini Project 2
Distributed Training for Healthcare Time-Series Models
Problem Statement
Implement distributed training approaches (data parallelism or pipeline parallelism) for the ECG/EEG classification model developed in Mini Project 1.
Scope (1 Month)
- Implement distributed training across nodes/devices
- Compare training performance with centralized model

Evaluation
- Training speed

- Memory usage
- System scalability

Expected Outcomes

- Understanding distributed training for healthcare time-series models

Output

✓ Distributed ECG/EEG model implementation

✓ Performance comparison report

Mini Project 3

Federated Learning for Wearable Healthcare Data

Problem Statement

Develop a federated learning system where wearable devices collaboratively train an ECG/EEG classification model without sharing raw data.

Scope (1 Month)

- Implement federated learning using Flower or PySyft
- Simulate multiple wearable devices/clients
- Perform federated aggregation

Evaluation

- Communication rounds
- Model convergence
- Client participation

Expected Outcomes

- Privacy-preserving wearable healthcare analytics

Final Output (Major Project 2)

✓ Federated ECG/wearable sensor classification system

✓ Performance comparison between centralized, distributed, and federated training